

LoRA-Adapted CLIP for Prompt-Based Multi-Label Finding Classification in Four-View Mammography

Heidi A. Holiel^{1,*}, Essam Rashed^{2,3}, Nancy Salem¹, Rana Hossam-Eldin¹

¹Biomedical Engineering Department, Capital University, Cairo, Egypt

²Graduate School of Information Science, University of Hyogo, Kobe 650-0047, Japan

³Advanced Medical Engineering Research Institute, University of Hyogo, Himeji 670-0836, Japan

* ORCID: 0009-0002-6452-5052

Abstract—A resource-efficient adaptation of CLIP is presented for prompt-based multi-label finding classification in four-view screening mammography. The method applies low-rank adaptation (LoRA) to the text encoder while keeping the vision encoder frozen and combines whole-view and lesion-level region-of-interest (ROI) contrastive objectives to adapt general vision-language representations to mammographic findings. Experiments on the 1,000-study held-out test set of VinDr-Mammo show that the best evaluated configuration achieves a support-filtered macro area under the receiver operating characteristic curve (AUC) of 0.637 using max pooling across the four views, while capturing 49.4% of finding-positive studies when the top 30% of studies are flagged for radiologist review. We further examine severe multi-label class imbalance and report a support-filtered macro-AUC together with per-category metrics and bootstrap confidence intervals to avoid unstable aggregate estimates from near-universal and low-support categories. Increasing the relative weight of the ROI objective produced the highest observed point estimate, but its effect cannot be separated from the additional training exposure introduced by warm-started continuation. Overall, the results demonstrate a parameter-efficient approach for adapting general-purpose vision-language models to resource-constrained medical imaging while providing a deployment-oriented triage evaluation.

Index Terms—vision-language models, CLIP, low-rank adaptation, mammography, computer-aided diagnosis, class imbalance, human-in-the-loop, prompt-based classification.

I. INTRODUCTION

Screening mammography is the main imaging tool for early breast cancer detection. Its interpretation is limited by substantial inter-reader variability and non-trivial false-positive and false-negative rates [1], and growing evidence suggests that AI can help. The MASAI randomised controlled trial has since shown that AI-supported single reading can match standard double reading while substantially reducing radiologist workload [2]. That evidence repositions the practical goal: rather than replacing the radiologist, AI should act as a triage layer that ranks studies by suspicion and routes the highest-risk cases for priority review.

Most current AI mammography systems are closed-set discriminative classifiers trained on a fixed label vocabulary, which limits their adaptability when clinical terminology changes or new finding types need to be added. A different approach is to align images and free-text descriptions in a shared embedding space and score images against arbitrary text prompts at inference time, the principle underlying CLIP [3].

Because no classification head is needed, new categories can be added by writing a new prompt, without any retraining. The approach has proven effective in chest radiology and in general biomedical imaging [4], [5]. Comparatively little work has examined CLIP adaptation for multi-view screening mammography, where each study spans four anatomically distinct views and where clinically important findings are often small, visually subtle, and severely underrepresented in annotated datasets.

Adapting a large pretrained model to a relatively small domain-specific dataset motivates parameter-efficient methods. We therefore apply LoRA [6] to the text encoder while keeping the vision encoder frozen, and hold this adaptation strategy fixed while varying the relative weighting of the two contrastive objectives. We train a LoRA-adapted CLIP model for four-view mammography on VinDr-Mammo [7] using paired per-view and ROI contrastive objectives. We evaluate study-level triage recall, introduce a support-filtered macro AUC for severe multi-label imbalance, and investigate whether increasing the relative weight of the ROI objective improves discrimination of small findings.

II. RELATED WORK

Convolutional networks trained at scale have matched or exceeded individual radiologist performance on mammography cancer detection retrospectively, with demonstrated cross-population generalisation [1] and multi-view patch-to-breast pipelines validated against radiologist readers [8]; class imbalance and limited labelled data remain the persistent obstacles [9]. Prospectively, the MASAI trial [2] randomised over 100,000 participants in the Swedish screening programme and found AI-supported single reading non-inferior to double reading, with a 44% reduction in radiologist workload and a 29% relative increase in detected cancers. This motivates evaluating against the metric that matters in deployment, how many confirmed findings are captured at a given flagging rate, which we adopt directly. Our work departs from the discriminative line in using prompt-based vision-language alignment rather than a fixed classification head.

CLIP [3] learns a joint image-text embedding space enabling prompt-based classification via prompts. General CLIP transfers poorly to biomedical images, which BiomedCLIP addressed through large-scale biomedical caption pretraining [5].

TABLE I
TEST-SET ($n = 1,000$ STUDIES) POSITIVE COUNTS BY FINDING
CATEGORY. CATEGORIES ARE NOT MUTUALLY EXCLUSIVE.

Category	Positives (n)	Prevalence (%)
Mass	115	11.5
Suspicious Calcification	54	5.4
Architectural Distortion	13	1.3
Asymmetry	20	2.0
Focal Asymmetry	26	2.6
Global Asymmetry	3	0.3
No Finding	997	99.7

In chest radiology, CheXzero [4] showed that contrastive training on chest X-ray and report pairs, without explicit pathology labels, yields prompt-based AUC comparable to board-certified radiologists. We apply this principle to multi-view mammography, which has more complex acquisition geometry, fewer public image-annotation pairs, and lesions smaller relative to image area. Adaptation uses LoRA [6], whose low-rank constraint acts as an implicit regulariser suited to limited labelled data. On evaluation, Rajaraman et al. [10] examined imbalance-calibration-threshold interactions, and Litjens et al. [9] note that naive aggregates misrepresent per-category performance under imbalance; we extend this specifically to multi-label macro AUC.

III. MATERIALS AND METHODS

A. Dataset

VinDr-Mammo [7] is a public full-field digital mammography dataset of 5,000 four-view studies (20,000 images) from two Vietnamese hospitals, independently double-read with disagreements resolved by a third radiologist. Each study provides bilateral CC and MLO views, breast-level BI-RADS assessment, breast density rating, and bounding-box annotations for each finding. We follow the standard partition of 4,000 training and 1,000 held-out test studies, and carve an internal validation set of 600 studies (15%) from the training partition, sampled at the study level to prevent leakage. The test set is not accessed for any purpose until final evaluation.

Seven finding categories are used: Mass, Suspicious Calcification, Architectural Distortion, Asymmetry, Focal Asymmetry, Global Asymmetry, and No Finding. Four rare descriptors (Nipple Retraction, Skin Retraction, Skin Thickening, Suspicious Lymph Node) were excluded for insufficient positive support; extending coverage to these is left to future work. Table I summarises the test-set positive counts.

The class imbalance is severe. No Finding appears in 99.7% of test studies, only two finding categories exceed fifty positives, and Global Asymmetry is present in just three.

Lesions are generally small relative to the full mammographic view: median bounding-box area is 0.41% for suspicious calcification and 0.58% for mass. This motivates the ROI-level objective in Section III-B, while Fig. 1 shows the finding distribution and bounding-box areas; representative annotated crops are shown in Fig. 2.

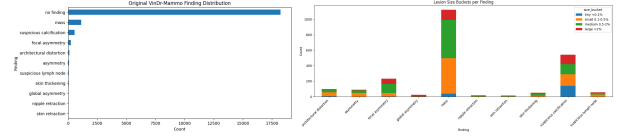


Fig. 1. Left: image-level finding distribution in VinDr-Mammo. Right: bounding-box area as a fraction of image area by finding category.

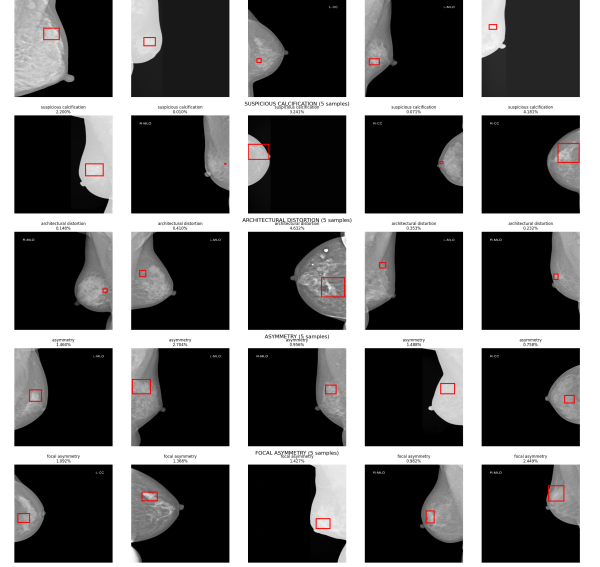


Fig. 2. Representative annotated examples (red boxes) for the five clinically relevant finding categories, with bounding-box area shown as a percentage of image area. Many lesions are visually subtle and lesion size varies substantially within each category.

Findings are also unevenly distributed across the two projections. Asymmetry annotations, for instance, appear roughly three times as often on MLO as on CC (69 versus 22), while Mass and Suspicious Calcification are close to balanced. A finding is therefore often annotated on one projection of a breast and not the other, which is the basis for the max-pooling aggregation compared in Section III-B.

B. Model Architecture

The base model is CLIP with a ViT-B/16 image encoder and a transformer text encoder, both initialised from OpenAI’s general-domain pretrained weights [3]. LoRA rank-8 adapters are applied to the text encoder only: trainable low-rank matrices ($r = 8$, $\alpha = 16$, dropout 0.05) are inserted into the query, key, value, and output projection layers as well as both feed-forward layers of each of the 12 transformer blocks in the text encoder [6]. The vision encoder is kept frozen at its pretrained weights throughout all training; only the LoRA parameters and a learned temperature scalar τ are updated. We keep the vision encoder frozen to preserve the pretrained representation and isolate the effect of the contrastive loss weighting across training configurations.

For prompt-based study-level scoring, only the four full views are encoded. Bounding-box crops are used during train-

ing and are never required at inference, so no lesion annotation is needed on a test study. Two aggregation strategies over the per-view embeddings are compared. *Mean pooling* averages the four per-view image embeddings before computing cosine similarity against a text prompt. *Max pooling* scores each view independently and retains the highest similarity across the four views. Max pooling is motivated by the observation that a finding often appears in only one of the four views, for example, a mass visible on the MLO projection but occult on the CC of the same breast, so averaging may dilute a genuine single-view signal.

Training uses a symmetric InfoNCE contrastive loss at two spatial granularities. The *per-view loss* contrasts each resized full mammographic view against a text caption encoding its laterality, projection, and annotated findings. The *ROI loss* contrasts bounding-box crops pre-resized to 336×336 pixels against the same captions. This crop-level objective is included precisely because the small lesions that are most clinically important lose spatial detail when full views are downsampled to the encoder’s 224×224 input: at that resolution, a suspicious calcification covering 0.3% of the image occupies roughly 13×13 pixels. The ROI crop, by targeting the lesion region directly before encoding, preserves far more of the relevant texture. The relative weight of the two terms, *view_weight* and *roi_weight*, is treated as a training hyperparameter and is the single training variable manipulated in this study. The ROI term acts as a training-time signal only: any benefit it confers must survive the transfer from crop-level supervision to full-view scoring at test time.

C. Training Protocol and Checkpoints

Three checkpoints from a single training lineage are evaluated. *lora* (epoch 16) is the best-validation checkpoint from the baseline LoRA run, trained on 3,400 studies with (*view_weight*=1.0, *roi_weight*=0.5). *lora_final* (epoch 23) is the final checkpoint from the same run and serves as the warm-start for *reweighted*. The latter continues training with inverted loss weights (*view_weight*=0.5, *roi_weight*=1.5). Thus, the backbone, data split, and LoRA adaptation strategy remain fixed across the lineage, while the *reweighted* continuation changes the relative emphasis of the two contrastive objectives.

The *lora* checkpoint was selected by minimum validation loss (epoch 16, 3.015), whereas *lora_final* corresponds to epoch 23, after validation loss had continued to improve and then plateaued. These are not independent experiments but two checkpoints from the same run. The *reweighted* checkpoint is motivated by the hypothesis that increasing the ROI-loss contribution provides a stronger training signal for localized, fine-grained findings that may be poorly represented after whole-view downsampling. All runs use AdamW, automatic mixed precision, and gradient-norm clipping at 1.0. The *reweighted* continuation uses a learning rate of 2×10^{-5} , weight decay of 1×10^{-4} , and early stopping with patience 4 based on validation loss. The No Finding subsample is redrawn at the start of each training epoch to expose the

TABLE II
PROMPT-BASED TEMPLATES. $\{\text{FINDING}\} \rightarrow \text{LOWER-CASED CATEGORY NAME}$. NO FINDING SUBSTITUTES FIXED PHRASES FOR TEMPLATES 1–3.

#	Template
1	Mammogram showing {finding}.
2	A breast mammogram with {finding}.
3	Radiology report: {finding} identified.
4	Finding: {finding}.
<i>No Finding only (templates 1–3 with fixed phrases):</i>	
	Mammogram showing no abnormal findings.
	Mammogram showing no significant findings.
	Mammogram showing unremarkable study.

model to greater diversity within the majority class. For the reweighted continuation, training loss decreased throughout the eight epochs, while validation loss reached its minimum at epoch 4; this checkpoint is therefore used for evaluation. The complete loss trajectories are shown in Fig. 3.

An important limitation is that *reweighted* differs from *lora_final* in both loss weights and cumulative training exposure because it is warm-started. Therefore, any performance difference cannot be attributed to ROI-loss reweighting alone. A matched-epoch continuation from *lora_final* that retains the original loss weights would be required for a causal comparison and remains the most important follow-up experiment.

D. Evaluation Protocol

All evaluation is performed on the 1,000-study held-out test set. Prompt-based multi-label classification encodes each finding category using an ensemble of text prompt templates (Table II) and scores each study’s image embedding(s) by cosine similarity, converted to a pseudo-probability using a temperature-scaled sigmoid.

a) Support-filtered macro AUC.: We report the mean per-category AUC across finding categories with at least 20 positive test instances, excluding No Finding from the primary aggregate. Two independent distortions motivate this: No Finding, present in 99.7% of studies, dominates an unweighted average regardless of minority-category performance, and categories below twenty positives yield intervals too wide to be informative in isolation. Architectural Distortion (13 positives) and Global Asymmetry (3 positives) are therefore reported individually but excluded from the aggregate. Statistical uncertainty is estimated using percentile bootstrap resampling (1,000 resamples) with 95% confidence intervals. Overlapping intervals are reported without claiming a superior checkpoint.

b) Triage performance.: Studies are ranked by predicted score for each finding category, and the top k fraction is flagged. Recall at flagging rate k is the proportion of true positives captured. We report recall at 30% and the full 5–50% flagging-rate curve, following the triage framing of the MASAI trial [2].

c) Saliency maps.: Because the dual-encoder model produces image–text similarity rather than class-conditioned log-

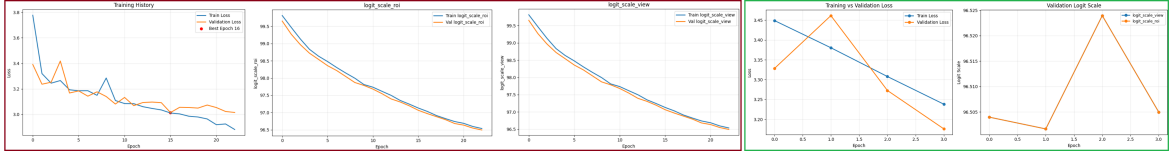


Fig. 3. Training and validation losses for the initial LoRA run and its warm-started reweighted continuation. The epoch-23 `lora_final` checkpoint initializes the reweighted run, and the epoch-4 checkpoint, corresponding to minimum validation loss, is used for evaluation.

TABLE III

SUPPORT-FILTERED MACRO AUC BY CHECKPOINT AND AGGREGATION STRATEGY. CHECKPOINTS ARE ORDERED BY CUMULATIVE TRAINING EXPOSURE. BOOTSTRAP 95% CIs (1,000 RESAMPLES) ARE SHOWN FOR MEAN-POOL.

Checkpoint	Mean-Pool	Max-Pool	95% CI
<code>lora</code> (ep. 16)	0.588	0.569	[0.535, 0.643]
<code>lora_final</code> (ep. 23)	0.609	0.614	[0.555, 0.669]
<code>reweighted</code>	0.623	0.637	[0.566, 0.680]

its, we use a sliding-window similarity method. Overlapping 224×224 windows are extracted from a resized 800×800 image at a 64-pixel stride. For each window, CLIP similarity to a finding-specific prompt is computed relative to a fixed negative prompt describing normal fibroglandular tissue. The resulting difference maps reduce global similarity bias and are used as qualitative diagnostics only; they are not validated against radiologist-annotated bounding boxes.

IV. RESULTS

Table III reports the support-filtered macro AUC for the three checkpoints under both aggregation strategies. The reweighted checkpoint achieves the highest point estimate under both mean pooling (0.623) and max pooling (0.637), followed by `lora_final` (0.609/0.614) and `lora` (0.588/0.569).

The ordering is monotone under both pooling strategies, but two constraints bound what it establishes. All three bootstrap intervals overlap, the reweighted interval ([0.566, 0.680]) contains both earlier point estimates, so no pairwise difference is statistically separable at this test-set size. And the ordering coincides exactly with cumulative training exposure, which is confounded with the loss-weight change by construction. We therefore report the reweighted checkpoint as having the best point estimate, without claiming a demonstrated benefit from ROI upweighting. Max pooling exceeds mean pooling for the two later checkpoints (+0.005, +0.014) but not for `lora` (−0.019). The differences are small relative to the intervals; max pooling is used for detailed reporting on the basis of its best single point estimate.

Table IV and Fig. 4 report per-category AUC and PR-AUC for the best checkpoint (reweighted, max pooling). Among reliably-supported categories, Suspicious Calcification achieved the highest AUC (0.676), followed by Asymmetry (0.649), Focal Asymmetry (0.636), and Mass (0.586). The low PR-AUC values for all finding categories (0.037–0.216)

TABLE IV

PER-CATEGORY PERFORMANCE (REWEIGHTED CHECKPOINT, MAX POOLING). CATEGORIES WITH < 20 POSITIVES ARE EXCLUDED FROM THE SUPPORT-FILTERED MACRO AUC. PR-AUC IS LOW FOR MINORITY CATEGORIES DUE TO CLASS IMBALANCE.

Category	n	AUC	PR-AUC	Reliable
Mass	115	0.586	0.216	Yes
Susp. Calcification	54	0.676	0.160	Yes
Arch. Distortion	13	0.544	0.016	No
Asymmetry	20	0.649	0.037	Yes
Focal Asymmetry	26	0.636	0.050	Yes
Global Asymmetry	3	0.400	0.003	No
No Finding	997	0.034	0.990	—

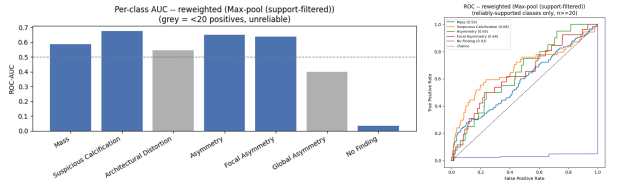


Fig. 4. ROC curves for reliably-supported categories plus No Finding, whose substantially below-chance behaviour (AUC 0.034) illustrates why it is excluded from any aggregate. Per-category AUC values are given in Table IV.

are expected under this level of imbalance: at prevalences of 2–12%, even a strong discriminator achieves low average precision because the operating region of interest lies at very high recall. No Finding achieved a PR-AUC of 0.990, driven entirely by base rate rather than discriminability.

Table V and Fig. 5 report recall at the 30% flagging rate. Averaged across the four reliably-supported categories, mean recall is 0.494: flagging fewer than one-third of all studies captures roughly half of those containing a confirmed finding. Per-category recall ranged from 38.3% for Mass to 59.3% for Suspicious Calcification.

V. DISCUSSION

The checkpoint comparison yields one clear methodological finding and one directional empirical one. The clear finding concerns measurement rather than modeling. Under naive macro-averaging the gaps between checkpoints are dominated by No Finding and the low-support categories, which vary erratically and obscure the categories with greater clinical relevance; under the support-filtered metric the ordering stabilises and becomes interpretable. This holds regardless of which checkpoint is preferred. The reweighted checkpoint achieved the best point estimate under both aggregation strategies and

TABLE V
RECALL AT A 30% STUDY-FLAGGING RATE (REWEIGHTED, MAX-POOL CHECKPOINT). MEAN RECALL ACROSS RELIABLY-SUPPORTED CATEGORIES IS 49.4%.

Category	n	Recall @ 30%
Mass	115	0.383
Suspicious Calcification	54	0.593
Arch. Distortion*	13	0.385
Asymmetry	20	0.500
Focal Asymmetry	26	0.500
Global Asymmetry*	3	0.000

Mean (reliable)

0.494

* Below reliability threshold; included for completeness only.

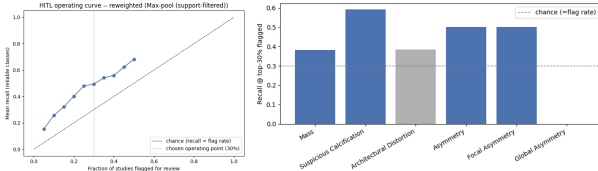


Fig. 5. Mean recall (reliably-supported categories) versus study-flagging rate, 5% to 50%. The curve stays above the chance diagonal (recall = flagging rate) at every tested point, confirming discriminative signal beyond random ordering. The 30% operating point of Table V is marked.

improved the point estimate for every adequately supported finding category. However, overlapping confidence intervals and the additional training exposure prevent a causal attribution to ROI reweighting. A matched-epoch control is therefore required.

Holding the adaptation strategy fixed at LoRA means this study makes no empirical claim about LoRA relative to full-parameter fine-tuning; that choice rests on prior literature [6] and corpus size, not on evidence generated here.

The absolute AUC range of 0.59–0.64 is substantially below CheXzero’s performance on chest pathologies [4]. The gap is most plausibly explained by data scale: CheXzero and BiomedCLIP trained on hundreds of thousands to fifteen million image-text pairs [4], [5], while this work adapts CLIP from roughly 3,400 training studies. The methodological improvements described here reduce metric distortion and improve the training signal, but they are unlikely to close a four-orders-of-magnitude data gap. Scaling the training corpus, through additional annotated mammography datasets or weakly supervised pretraining on mammography radiology reports, is the more direct lever on absolute performance.

The No Finding AUC of 0.034 warrants comment but little interpretation. With 997 positives among 1,000 test studies, the estimate rests on three negative studies, and is therefore subject to precisely the low-support instability our support filter is designed to exclude; our reliability criterion counts positive instances only, and No Finding fails the equivalent test on the negative side. We report the value for completeness and exclude it from every aggregate. The direction is nonetheless consistent with a known weakness of CLIP text encoders, which represent negation poorly: a prompt describing the

absence of findings may embed near the finding prompts rather than opposite them, so abnormal studies score high on both. A positively phrased normal-tissue prompt, of the kind already used as the negative reference in our saliency analysis, provides a straightforward and testable direction for future work.

Capturing 49.4% of studies with a confirmed finding while flagging only 30% of the screening population for review demonstrates discriminative signal above the random-flagging baseline, and the recall curve remaining above the diagonal across the full 5–50% range confirms this is not an artifact of threshold selection. Whether this level of recall is sufficient for a clinical triage application depends on the tradeoff a deploying institution is willing to accept, which is why we report the full curve rather than a single number. Compared with the MASAI trial [2], where AI-supported single reading matched double reading, our system is at an earlier stage of development; it is a proof of concept for the prompt-based approach on a small labelled corpus, not a system ready for clinical deployment.

The study is limited by its use of a single dataset from two Vietnamese hospitals, so generalisability to other populations, vendors, and imaging protocols is unknown. Architectural Distortion (13 positives) and Global Asymmetry (3) have sparse support and are therefore exploratory only. The reweighting comparison is also confounded because *reweighted* differs from *lora_final* in both loss weights and training steps; consequently, checkpoint rankings are point estimates pending a matched-epoch control. Finally, the study does not compare LoRA with full-parameter fine-tuning or a closed-set convolutional baseline on the identical split, and its retrospective design provides no prospective clinical evidence.

VI. CONCLUSION

We have trained and evaluated a prompt-based multi-label mammography finding classifier by applying LoRA to the text encoder of CLIP and fine-tuning with paired per-view and ROI contrastive objectives on VinDr-Mammo. The best checkpoint reaches a support-filtered macro AUC of 0.637 and a mean triage recall of 49.4% when 30% of studies are flagged for review, exceeding a random-flagging baseline at every operating point evaluated and across all finding categories with adequate test-set support.

Two methodological contributions accompany this result. A support-filtered macro AUC excluding a dominant negative category and low-support rare classes gives a more stable aggregate than naive macro-averaging, and changes which checkpoint appears best. Upweighting the ROI-level loss improves the point-estimate AUC directionally across every finding category, though separating this from the warm-started run’s additional training exposure requires the matched-epoch control that remains the immediate next experiment. The primary bottleneck for future improvement is data scale: closing the gap with large-scale chest radiograph vision-language models needs a larger and more diverse mammography corpus more than it needs further methodological refinement.

REFERENCES

- [1] S. M. McKinney, M. Sieniek, V. Godbole, J. Godwin, N. Antropova, H. Ashrafian, T. Back, M. Chesus, G. C. Corrado, A. Darzi, M. Etemadi, F. Garcia-Vicente, F. J. Gilbert, M. Halling-Brown, D. Hassabis, S. Jansen, A. Karthikesalingam, C. J. Kelly, D. King, J. R. Ledsam, D. Melnick, H. Mostofi, L. Peng, J. J. Reicher, B. Romera-Paredes, R. Sidebottom, M. Suleyman, D. Tse, K. C. Young, J. De Fauw, and S. Shetty, "International evaluation of an AI system for breast cancer screening," *Nature*, vol. 577, pp. 89–94, Jan. 2020.
- [2] K. Lång, V. Josefsson, A.-M. Larsson, S. Larsson, C. Högberg, H. Sartor, S. Hofvind, I. Andersson, and A. Rosso, "Artificial intelligence-supported screen reading versus standard double reading in the Mammography Screening with Artificial Intelligence trial (MASAI): a clinical safety analysis of a randomised, controlled, non-inferiority, single-blinded, screening accuracy study," *The Lancet Oncology*, vol. 24, pp. 936–944, Aug. 2023.
- [3] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, vol. 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763, 2021.
- [4] E. Tiu, E. Talus, P. Patel, C. P. Langlotz, A. Y. Ng, and P. Rajpurkar, "Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning," *Nature Biomedical Engineering*, vol. 6, pp. 1399–1406, Dec. 2022.
- [5] S. Zhang, Y. Xu, N. Usuyama, H. Xu, J. Bagga, R. Tinn, S. Preston, R. Rao, M. Wei, N. Valluri, C. Wong, A. Tupini, Y. Wang, M. Mazzola, S. Shukla, L. Liden, J. Gao, M. P. Lungren, T. Naumann, S. Wang, and H. Poon, "A multimodal biomedical foundation model trained from fifteen million image-text pairs," *NEJM AI*, 2024. Also available as arXiv:2303.00915.
- [6] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.
- [7] H. T. Nguyen, H. Q. Nguyen, H. H. Pham, K. Lam, L. T. Le, M. Dao, and V. Vu, "VinDr-Mammo: A large-scale benchmark dataset for computer-aided diagnosis in full-field digital mammography," *Scientific Data*, vol. 10, May 2023.
- [8] N. Wu, J. Phang, J. Park, Y. Shen, Z. Huang, M. Zorin, S. Jastrzebski, T. Févry, J. Katsnelson, E. Kim, S. Wolfson, U. Parikh, S. Gaddam, L. L. Y. Lin, K. Ho, J. D. Weinstein, B. Reig, Y. Gao, H. Toth, K. Pysarenko, A. Lewin, J. Lee, K. Airola, E. Mema, S. Chung, E. Hwang, N. Samreen, S. G. Kim, L. Heacock, L. Moy, K. Cho, and K. J. Geras, "Deep neural networks improve radiologists' performance in breast cancer screening," *IEEE Transactions on Medical Imaging*, vol. 39, pp. 1184–1194, Apr. 2020.
- [9] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, Dec. 2017.
- [10] S. Rajaraman, P. Ganesan, and S. Antani, "Deep learning model calibration for improving performance in class-imbalanced medical image classification tasks," *PLoS ONE*, vol. 17, p. e0262838, Jan. 2022.